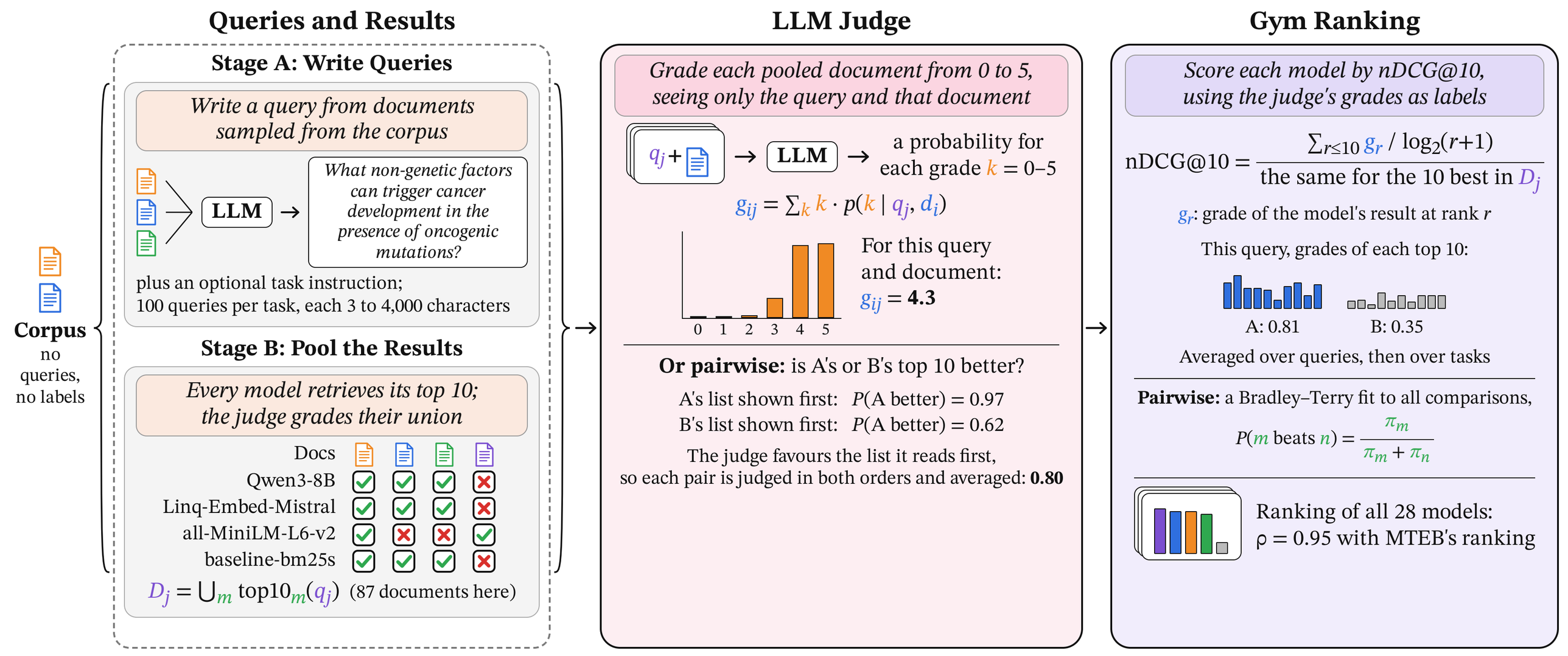

Queries and Results
LLM Judge
Gym Ranking
Corpus
no queries, no labels
Stage A: Write Queries
Write a query from documents sampled from the corpus
LLM
What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?
plus an optional task instruction;
100 queries per task, each 3 to 4,000 characters
Stage B: Pool the Results
Every model retrieves its top 10; the judge grades their union
Docs
Qwen3-8B
Linq-Embed-Mistral
all-MiniLM-L6-v2
baseline-bm25s
$D_j = \bigcup_m \text{top10}_m(q_j)$ (87 documents here)
Grade each pooled document from 0 to 5, seeing only the query and that document
q_j + → LLM → a probability for each grade k = 0–5
$g_{ij} = \sum_k k \cdot p(k \mid q_j, d_i)$
0 1 2 3 4 5
For this query and document: g_{ij} = 4.3
Or pairwise: is A's or B's top 10 better?
A's list shown first: P(A better) = 0.97
B's list shown first: P(A better) = 0.62
The judge favours the list it reads first, so each pair is judged in both orders and averaged: 0.80
Score each model by nDCG@10, using the judge's grades as labels
$\text{nDCG@10} = \frac{\sum_{r \le 10} g_r / \log_2(r+1)}{\text{the same for the 10 best in } D_j}$
g_r: grade of the model's result at rank r
This query, grades of each top 10:
A: 0.81
B: 0.35
Averaged over queries, then over tasks
Pairwise: a Bradley–Terry fit to all comparisons,
$P(m \text{ beats } n) = \frac{\pi_m}{\pi_m + \pi_n}$
Ranking of all 28 models:
ρ = 0.95 with MTEB's ranking